

Answers

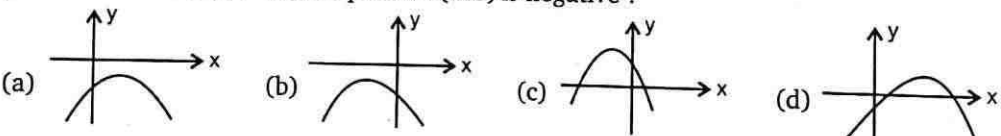
1. (d)	2. (c)	3. (a)	4. (b)	5. (a)	6. (a)	7. (b)	8. (c)	9. (a)	10. (d)
11. (c)	12. (c)	13. (a)	14. (a)	15. (d)	16. (d)	17. (b)	18. (b)	19. (c)	20. (a)
21. (b)	22. (d)	23. (c)	24. (a)	25. (d)	26. (a)	27. (d)	28. (c)	29. (d)	30. (c)
31. (d)	32. (d)	33. (c)	34. (d)	35. (a)	36. (c)	37. (b)	38. (b)	39. (c)	40. (c)
41. (b)	42. (b)	43. (c)	44. (d)	45. (b)	46. (c)	47. (a)	48. (a)	49. (c)	50. (c)
51. (b)	52. (b)	53. (a)	54. (a)	55. (c)	56. (b)	57. (c)	58. (a)	59. (c)	60. (a)
61. (c)	62. (a)	63. (a)	64. (c)	65. (a)	66. (c)	67. (a)	68. (d)	69. (b)	70. (d)
71. (c)	72. (c)	73. (c)	74. (a)	75. (c)	76. (c)	77. (b)	78. (a)	79. (b)	80. (c)
81. (b)	82. (a)	83. (a)	84. (a)	85. (b)	86. (b)	87. (d)	88. (b)	89. (d)	90. (b)
91. (b)	92. (c)	93. (c)	94. (b)	95. (d)	96. (b)	97. (b)	98. (d)	99. (a)	100. (b)
101. (d)	102. (b)	103. (c)	104. (b)	105. (c)	106. (d)	107. (d)	108. (d)	109. (c)	110. (a)
111. (a)	112. (b)	113. (c)	114. (c)	115. (b)	116. (c)	117. (d)	118. (d)	119. (b)	120. (d)
121. (d)	122. (b)	123. (c)	124. (b)	125. (d)	126. (d)	127. (c)	128. (b)	129. (b)	130. (b)
131. (c)	132. (b)	133. (b)	134. (b)	135. (c)	136. (d)	137. (b)	138. (c)	139. (a)	140. (b)
141. (b)	142. (d)	143. (a)	144. (c)	145. (a)	146. (b)	147. (b)	148. (b)	149. (c)	150. (c)
151. (b)	152. (c)	153. (a)							

Exercise-2 : One or More than One Answer Is/are Correct

- Let S is the set of all real x such that $\frac{2x-1}{2x^3+3x^2+x}$ is positive, then S contains :
 (a) $\left(-\infty, -\frac{3}{2}\right)$ (b) $\left(-\frac{3}{2}, -\frac{1}{2}\right)$ (c) $\left(-\frac{1}{2}, 0\right)$ (d) $\left(\frac{1}{2}, 2\right)$
- If $kx^2 - 4x + 3k + 1 > 0$ for atleast one $x > 0$, then if $k \in S$, then S contains:
 (a) $(1, \infty)$ (b) $(0, \infty)$ (c) $(-1, \infty)$ (d) $\left(-\frac{1}{4}, \infty\right)$
- The equation $|x^2 - x - 6| = x + 2$ has:
 (a) two positive roots (b) two real roots
 (c) three real roots (d) four real roots
- If the roots of the equation $x^2 - ax - b = 0$ ($a, b \in \mathbb{R}$) are both lying between -2 and 2 , then :
 (a) $|a| < 2 - \frac{b}{2}$ (b) $|a| > 2 - \frac{b}{2}$
 (c) $|a| < 4$ (d) $|a| > \frac{b}{2} - 2$
- Consider the equation in real number x and a real parameter λ , $|x-1| - |x-2| + |x-4| = \lambda$. Then for $\lambda \geq 1$, the number of solutions, the equation can have is/are :
 (a) 1 (b) 2 (c) 3 (d) 4
- If a and b are two distinct non-zero real numbers such that $a - b = \frac{a}{b} = \frac{1}{b} - \frac{1}{a}$, then :
 (a) $a > 0$ (b) $a < 0$ (c) $b < 0$ (d) $b > 0$
- Let $f(x) = ax^2 + bx + c$, $a > 0$ and $f(2-x) = f(2+x) \forall x \in \mathbb{R}$ and $f(x) = 0$ has 2 distinct real roots, then which of the following is true ?
 (a) Atleast one root must be positive
 (b) $f(2) < f(0) > f(1)$
 (c) Minimum value of $f(x)$ is negative
 (d) Vertex of graph of $y = f(x)$ lies in 3rd quadrat
- In the above problem, if roots of equation $f(x) = 0$ are non-real complex, then which of the following is false ?
 (a) $f(x) = \sin \frac{\pi x}{4}$ must have 2 solutions
 (b) $4a - 2b + c < 0$
 (c) If $\log_{f(2)} f(3)$ is not defined, then $f(x) \geq 1 \forall x \in \mathbb{R}$
 (d) All a, b, c are positive
- If exactly two integers lie between the roots of equation $x^2 + ax - 1 = 0$. Then integral value(s) of ' a ' is/are :
 (a) -1 (b) -2 (c) 1 (d) 2

10. If the minimum value of the quadratic expression $y = ax^2 + bx + c$ is negative attained at negative value of x , then :
 (a) $a > 0$ (b) $b > 0$ (c) $c > 0$ (d) $D > 0$
 (where D is discriminant)
11. The quadratic expression $ax^2 + bx + c > 0 \forall x \in R$, then:
 (a) $13a - 5b + 2c > 0$ (b) $13a - b + 2c > 0$
 (c) $c > 0, D < 0$ (d) $a + c > b, D < 0$
 (where D is discriminant)
12. The possible positive integral value of ' k ' for which $5x^2 - 2kx + 1 < 0$ has exactly one integral solution may be divisible by :
 (a) 2 (b) 3 (c) 5 (d) 7
13. If the equation $x^2 + px + q = 0$, the coefficient of x was incorrectly written as 17 instead of 13. Then roots were found to be -2 and -15. The correct roots are :
 (a) -1 (b) -3 (c) -5 (d) -10
14. If $x^2 - 3x + 2 > 0$ and $x^2 - 3x - 4 \leq 0$, then :
 (a) $|x| \leq 2$ (b) $2 \leq x \leq 4$ (c) $-1 \leq x < 1$ (d) $2 < x \leq 4$
15. If $5^x + (2\sqrt{3})^{2x} - 169 \leq 0$ is true for x lying in the interval :
 (a) $(-\infty, 2)$ (b) $(0, 2]$ (c) $(2, \infty)$ (d) $(0, 4)$
16. Let $f(x) = x^2 + ax + b$ and $g(x) = x^2 + cx + d$ be two quadratic polynomials with real coefficients and satisfy $ac = 2(b + d)$. Then which of the following is (are) correct?
 (a) Exactly one of either $f(x) = 0$ or $g(x) = 0$ must have real roots.
 (b) Atleast one of either $f(x) = 0$ or $g(x) = 0$ must have real roots.
 (c) Both $f(x) = 0$ and $g(x) = 0$ must have real roots.
 (d) Both $f(x) = 0$ and $g(x) = 0$ must have imaginary roots.
17. The expression $\frac{1}{\sqrt{x+2\sqrt{x-1}}} + \frac{1}{\sqrt{x-2\sqrt{x-1}}}$ simplifies to :
 (a) $\frac{2}{3-x}$ if $1 < x < 2$ (b) $\frac{2}{2-x}$ if $1 < x < 2$
 (c) $\frac{2\sqrt{x-1}}{(x-2)}$ if $x > 2$ (d) $\frac{2\sqrt{x-1}}{x+2}$ if $x > 2$
18. If all values of x which satisfies the inequality $\log_{(1/3)}(x^2 + 2px + p^2 + 1) \geq 0$ also satisfy the inequality $kx^2 + kx - k^2 \leq 0$ for all real values of k , then all possible values of p lies in the interval :
 (a) $[-1, 1]$ (b) $[0, 1]$ (c) $[0, 2]$ (d) $[-2, 0]$
19. Which of the following statement(s) is/are correct?
 (a) The number of quadratic equations having real roots which remain unchanged even after squaring their roots is 3.

- (b) The number of solutions of the equation $\tan 2\theta + \tan 3\theta = 0$, in the interval $[0, \pi]$ is equal to 6.
- (c) For $x_1, x_2, x_3 > 0$, the minimum value of $\frac{2x_1}{x_2} + \frac{128x_3^2}{x_2^2} + \frac{x_2^3}{4x_1x_3^2}$ equals 24.
- (d) The locus of the mid-points of chords of the circle $x^2 + y^2 - 2x - 6y - 1 = 0$, which are passing through origin is $x^2 + y^2 - x - 3y = 0$.
- 20.** If $(a, 0)$ is a point on a diameter inside the circle $x^2 + y^2 = 4$. Then $x^2 - 4x - a^2 = 0$ has :
- (a) Exactly one real root in $(-1, 0]$ (b) Exactly one real root in $[2, 5]$
 (c) Distinct roots greater than -1 (d) Distinct roots less than 5
- 21.** Let $x^2 - px + q = 0$ where $p \in R, q \in R, pq \neq 0$ have the roots α, β such that $\alpha + 2\beta = 0$, then:
- (a) $2p^2 + q = 0$ (b) $2q^2 + p = 0$ (c) $q < 0$ (d) $q > 0$
- 22.** If a, b, c are rational numbers ($a > b > c > 0$) and quadratic equation $(a + b - 2c)x^2 + (b + c - 2a)x + (c + a - 2b) = 0$ has a root in the interval $(-1, 0)$ then which of the following statement(s) is/are correct ?
- (a) $a + c < 2b$
 (b) both roots are rational
 (c) $ax^2 + 2bx + c = 0$ have both roots negative
 (d) $cx^2 + 2bx + a = 0$ have both roots negative
- 23.** For the quadratic polynomial $f(x) = 4x^2 - 8ax + a$, the statements(s) which hold good is/are:
- (a) There is only one integral 'a' for which $f(x)$ is non-negative $\forall x \in R$
 (b) For $a < 0$, the number zero lies between the zeroes of the polynomial
 (c) $f(x) = 0$ has two distinct solutions in $(0, 1)$ for $a \in \left(\frac{1}{7}, \frac{4}{7}\right)$
 (d) The minimum value of $f(x)$ for minimum value of a for which $f(x)$ is non-negative $\forall x \in R$ is 0
- 24.** Given a, b, c are three distinct real numbers satisfying the inequality $a - 2b + 4c > 0$ and the equation $ax^2 + bx + c = 0$ has no real roots. Then the possible value(s) of $\frac{4a + 2b + c}{a + 3b + 9c}$ is/are :
- (a) 2 (b) -1 (c) 3 (d) $\sqrt{2}$
- 25.** Let $f(x) = x^2 - 4x + c \forall x \in R$, where c is a real constant, then which of the following is/are true ?
- (a) $f(0) > f(1) > f(2)$ (b) $f(2) > f(3) > f(4)$
 (c) $f(1) < f(4) < f(-1)$ (d) $f(0) = f(4) > f(3)$
- 26.** If $0 < a < b < c$ and the roots α, β of the equation $ax^2 + bx + c = 0$ are imaginary, then :
- (a) $|\alpha| = |\beta|$ (b) $|\alpha| > 1$ (c) $|\beta| < 1$ (d) $|\alpha| = 1$
- 27.** If x satisfies $|x - 1| + |x - 2| + |x - 3| > 6$, then :
- (a) $x \in (-\infty, 1)$ (b) $x \in (-\infty, 0)$ (c) $x \in (4, \infty)$ (d) $(2, \infty)$

28. If both roots of the quadratic equation $ax^2 + x + b - a = 0$ are non real and $b > -1$, then which of the following is/are correct?
 (a) $a > 0$ (b) $a < b$ (c) $3a > 2 + 4b$ (d) $3a < 2 + 4b$
29. If a, b are two numbers such that $a^2 + b^2 = 7$ and $a^3 + b^3 = 10$, then :
 (a) The greatest value of $|a + b| = 5$ (b) The greatest value of $(a + b)$ is 4
 (c) The least value of $(a + b)$ is 1 (d) The least value of $|a + b|$ is 1
30. The number of non-negative integral ordered pair(s) (x, y) for which $(xy - 7)^2 = x^2 + y^2$ holds is greater than or equal to :
 (a) 1 (b) 2 (c) 3 (d) 4
31. If α, β, γ and δ are the roots of the equation $x^4 - bx - 3 = 0$; then an equation whose roots are $\frac{\alpha + \beta + \gamma}{\delta^2}, \frac{\alpha + \beta + \delta}{\gamma^2}, \frac{\alpha + \gamma + \delta}{\beta^2}$ and $\frac{\beta + \gamma + \delta}{\alpha^2}$ is :
 (a) $3x^4 + bx + 1 = 0$ (b) $3x^4 - bx + 1 = 0$
 (c) $3x^4 + bx^3 - 1 = 0$ (d) $3x^4 - bx^3 - 1 = 0$
32. The value of k for which both roots of the equation $4x^2 - 2x + k = 0$ are completely in $(-1, 1)$ may be equal to :
 (a) -1 (b) 0 (c) 2 (d) -3
33. If $a, b, c \in R$, then for which of the following graphs of the quadratic polynomial $y = ax^2 - 2bx + c$ ($a \neq 0$); the product (abc) is negative ?

34. If the equation $ax^2 + bx + c = 0$; $a, b, c \in R$ and $a \neq 0$ has no real roots then which of the following is/are always correct ?
 (a) $(a + b + c)(a - b + c) > 0$ (b) $(a + b + c)(a - 2b + 4c) > 0$
 (c) $(a - b + c)(4a - 2b + c) > 0$ (d) $a(b^2 - 4ac) > 0$
35. If α and β are the roots of the equation $ax^2 + bx + c = 0$; $a, b, c \in R$; $a \neq 0$ then which is (are) correct :
 (a) $\alpha^2 + \beta^2 = \frac{b^2 - 2ac}{a^2}$ (b) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{b^2 - 2ac}{c^2}$
 (c) $\frac{1}{\alpha^3} + \frac{1}{\beta^3} = \frac{abc - b^3}{c^3}$ (d) $\alpha\beta(\alpha + \beta) = \frac{-bc}{a^2}$
36. The equation $\cos^2 x - \sin x + \lambda = 0$, $x \in (0, \pi/2)$ has roots then value(s) of λ can be equal to :
 (a) 0 (b) -1 (c) 1/2 (d) 1
37. If the equation $\ln(x^2 + 5x) - \ln(x + a + 3) = 0$ has exactly one solution for x , then possible integral value of a is :
 (a) -3 (b) -1 (c) 0 (d) 2